CQF Exam One

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P.Task – The Global Minimum Variance Portfolio

Consider the asset universe below.

	Bonds	Equity	Real Estate	Commodities
Expected return μ_i	4%	8%	6%	10%
Standard deviation σ_i	5%	20%	12%	25%

The correlation matrix is

$$\rho = \begin{pmatrix} 1.00 & 0.20 & 0.15 & 0.05 \\ 0.20 & 1.00 & 0.60 & 0.40 \\ 0.15 & 0.60 & 1.00 & 0.30 \\ 0.05 & 0.40 & 0.30 & 1.00 \end{pmatrix}, \qquad r_f = 2\%.$$

Part I. Derivation of the GMV Portfolio

Derive the closed-form solution for the Global Minimum Variance (GMV) portfolio and compute its numerical weights.

(a)

Define the portfolio weight vector as

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{pmatrix},$$

where the four components correspond to bonds, equity, real estate, and commodities, respectively.

The expected-return vector and standard-deviation vector are

$$\boldsymbol{\mu} = \begin{pmatrix} 0.04 \\ 0.08 \\ 0.06 \\ 0.10 \end{pmatrix}, \qquad \boldsymbol{\sigma} = \begin{pmatrix} 0.05 \\ 0.20 \\ 0.12 \\ 0.25 \end{pmatrix}.$$

The expected return of the portfolio is

$$\mu_\Pi = \mathbf{w}^T \boldsymbol{\mu},$$

and the portfolio variance is

$$\sigma_\Pi^2 = \mathbf{w}^T \Sigma \mathbf{w},$$

where Σ is the covariance matrix.

The covariance matrix is obtained from the correlation matrix ρ using

$$\Sigma = S\rho S,$$

which is the covariance construction used in the CQF portfolio-optimisation notes (see [1], p. 17), where S is the diagonal matrix containing the standard deviations,

$$S = \text{diag}\,(0.05,\, 0.20,\, 0.12,\, 0.25) = \begin{pmatrix} 0.05 & 0 & 0 & 0 \\ 0 & 0.20 & 0 & 0 \\ 0 & 0 & 0.12 & 0 \\ 0 & 0 & 0 & 0.25 \end{pmatrix}.$$

The Global Minimum Variance portfolio is obtained by minimising the portfolio variance subject to the condition that the portfolio is fully invested. Therefore, the optimisation problem is

$$\boxed{\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w}}$$

subject to

$$\boxed{\mathbf{1}^T \mathbf{w} = 1}$$

or equivalently,

$$\sum_{i=1}^{4} w_i = 1.$$

The factor of $1/2$ does not affect the location of the minimum, but is included to simplify the derivative in the following derivation.

(b)

The GMV portfolio minimises the portfolio variance subject to the fully-invested constraint. Since multiplication of the objective function by a positive constant does not change the location of the minimum,

$$\min(\sigma_\Pi^2) = \min\left(\frac{1}{2}\sigma_\Pi^2\right).$$

The Lagrangian is therefore

$$\mathcal{L} = \frac{1}{2}\mathbf{w}^T\Sigma\mathbf{w} + \gamma(1 - \mathbf{1}^T\mathbf{w}),$$

where γ is the Lagrange multiplier associated with the fully-invested constraint.

The first-order condition is

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \Sigma\mathbf{w} - \gamma\mathbf{1} = 0.$$

Therefore,

$$\mathbf{w} = \gamma\Sigma^{-1}\mathbf{1}.$$

Substituting this result into the constraint $\mathbf{1}^T\mathbf{w} = 1$ gives

$$\begin{aligned}\mathbf{1}^T\left(\gamma\Sigma^{-1}\mathbf{1}\right) &= 1,\\ \gamma\mathbf{1}^T\Sigma^{-1}\mathbf{1} &= 1.\end{aligned}$$

Hence,

$$\gamma = \frac{1}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}}.$$

The closed-form expression for the Global Minimum Variance portfolio is therefore

$$\boxed{\mathbf{w}_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}}}.$$

This agrees with the minimum-variance formulation in the CQF optimisation notes (see [1], pp. 81–84).

(c)

Compute the allocations for the given universe of four assets, specific to this task.

Solution

Using the given asset standard deviations and correlation matrix, the covariance matrix is

$$\Sigma = \begin{pmatrix} 0.0025 & 0.002 & 0.0009 & 0.000625 \\ 0.002 & 0.04 & 0.0144 & 0.02 \\ 0.0009 & 0.0144 & 0.0144 & 0.009 \\ 0.000625 & 0.02 & 0.009 & 0.0625 \end{pmatrix}.$$

Using the closed-form GMV solution,

$$\mathbf{w}_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}},$$

we obtain

$$\Sigma^{-1}\mathbf{1} = \begin{pmatrix} \frac{27900}{71} \\ -\frac{3025}{142} \\ \frac{76375}{1278} \\ \frac{730}{71} \end{pmatrix}, \qquad \mathbf{1}^T\Sigma^{-1}\mathbf{1} = \frac{282245}{639}.$$

Therefore,

$$\mathbf{w}_{\text{GMV}} = \frac{639}{282245}\begin{pmatrix} \frac{27900}{71} \\ -\frac{3025}{142} \\ \frac{76375}{1278} \\ \frac{730}{71} \end{pmatrix} = \begin{pmatrix} \frac{50220}{56449} \\ -\frac{5445}{112898} \\ \frac{15275}{112898} \\ \frac{1314}{56449} \end{pmatrix}.$$

Numerically,

$$\mathbf{w}_{\text{GMV}} \approx \begin{pmatrix} 0.8897 \\ -0.0482 \\ 0.1353 \\ 0.0233 \end{pmatrix}.$$

The weights satisfy the fully-invested constraint,

$$\mathbf{1}^T\mathbf{w}_{\text{GMV}} = 1.$$

The resulting allocations are summarised below.

Asset	Allocation
Bonds	88.97%
Equity	-4.82%
Real Estate	13.53%
Commodities	2.33%

Table 1: GMV portfolio allocations.

Part II. Why Prefer the GMV Portfolio

An AI portfolio assistant has produced analysis over this universe, claiming that it "minimises portfolio risk" with an expected return of

$$\mu_{\Pi} = 4.5\%$$

and a risk of

$$\sigma_{\Pi} = 4.89\%.$$

It is likely that the AI referred to some specific target return.

(a)

Compare the GMV portfolio's risk to the AI's claimed risk of 4.89%. Is the AI's claim accurate in regard that it minimised risk in total, unconditionally?

Solution

Using the GMV weights obtained in Part I, the expected portfolio return is

$$\mu_{\Pi} = \mathbf{w}_{\text{GMV}}^{T}\boldsymbol{\mu} \approx 0.04218 = 4.218\%.$$

The corresponding portfolio volatility is

$$\sigma_{\Pi} = \sqrt{\mathbf{w}_{\text{GMV}}^{T}\Sigma\mathbf{w}_{\text{GMV}}} = \sqrt{\frac{6390639}{2822450000}} \approx 0.0476 = 4.76\%.$$

This is lower than the AI-claimed portfolio volatility of 4.89%. Therefore, the AI portfolio does not minimise portfolio risk unconditionally and is not the Global Minimum Variance portfolio.

The AI portfolio has a higher expected return of 4.5%, compared with approximately 4.22% for the GMV portfolio, but this is achieved at the cost of higher volatility. This suggests that the AI result may correspond to a minimum-variance portfolio subject to a specified target return rather than the unconditional GMV portfolio.

(b)

Give at least two reasons why the GMV portfolio is preferred in practice even though its Sharpe ratio is not the highest.

Solution

The GMV portfolio has several practical advantages. First, its allocation depends only on the covariance matrix Σ and does not require estimates of the expected-return vector $\boldsymbol{\mu}$. Expected returns are generally difficult to estimate accurately, so avoiding them reduces the amount of uncertain input entering the optimisation.

Second, covariance and volatility estimates can usually be estimated more reliably from historical data than expected returns. As a result, the GMV portfolio can be more robust out of sample than portfolios whose optimal weights depend strongly on forecasts of future returns. The CQF optimisation notes also discuss the difficulty of using historical expected-return estimates when they may not represent future expectations (see [1], p. 108).

Another advantage is that the GMV portfolio directly targets the lowest achievable portfolio volatility under the fully-invested constraint. This is useful for investors whose main objective is capital preservation or risk control rather than maximising expected excess return.

Finally, the GMV construction does not require an investor to specify subjective expected-return forecasts or a particular level of risk aversion. This makes the optimisation relatively simple to interpret and reduces dependence on additional modelling assumptions.

(c)

Consider the comparison of the GMV portfolio to the one that maximises the Sharpe Ratio. Provide the name of the latter and its closed-form formula to compute allocations (no derivation).

In your own words, explain whether that portfolio and/or the GMV portfolio is more sensitive to estimation errors in expected return $\boldsymbol{\mu}$.

Solution

The portfolio that maximises the Sharpe ratio is called the tangency portfolio. The CQF portfolio theory notes identify it as the risky portfolio with the highest Sharpe ratio (see [2], p. 48), while the optimisation notes derive the fully invested allocation (see [1], p. 94). Its allocation is

$$\mathbf{w}_{\tan} = \frac{\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{B - A r_f},$$

where

$$A = \mathbf{1}^T \Sigma^{-1} \mathbf{1} = \frac{282245}{639}, \qquad B = \boldsymbol{\mu}^T \Sigma^{-1} \mathbf{1} = \frac{15871}{852}.$$

For $r_f = 0.02$,

$$B - A r_f = \frac{125167}{12780},$$

and

$$\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1}) = \begin{pmatrix} \frac{1009}{142} \\ \frac{79}{1136} \\ \frac{17105}{10224} \\ \frac{1343}{1420} \end{pmatrix}.$$

Therefore,

$$\mathbf{w}_{\tan} = \frac{\begin{pmatrix} \frac{1009}{142} \\ \frac{79}{1136} \\ \frac{17105}{10224} \\ \frac{1343}{1420} \end{pmatrix}}{\frac{125167}{12780}} \approx \begin{pmatrix} 0.7255 \\ 0.0071 \\ 0.1708 \\ 0.0966 \end{pmatrix}.$$

And we can check $0.7255 + 0.0071 + 0.1708 + 0.0966 = 1$.

The tangency portfolio depends explicitly on the estimated expected-return vector $\boldsymbol{\mu}$, whereas the GMV portfolio does not. Therefore, estimation errors in expected returns directly affect the tangency-portfolio weights but do not directly affect the GMV weights.

Consequently, the tangency portfolio is more sensitive to estimation errors in $\boldsymbol{\mu}$ and may produce less stable allocations when the estimated expected returns change.

P.Task – Monte-Carlo Exploration of Feasible Region

The Monte-Carlo simulation for this section was carried out in the accompanying Excel workbook.

Instead of computing optimal allocations by ready formula, conduct the following experiment.

Generate at least 700 random allocation sets, each a 4×1 vector, satisfying only

$$\mathbf{w}'\mathbf{1} = 1.$$

One possible way is to generate three numbers w_1, w_2, w_3, then compute the fourth allocation from the constraint. Allocations may be negative and do not need to be optimal.

For each allocation set, compute

$$\mu_\Pi = \mathbf{w}'\boldsymbol{\mu}$$

and

$$\sigma_\Pi = \sqrt{\mathbf{w}'\Sigma\mathbf{w}}.$$

Plot the points with μ_Π on the vertical axis and σ_Π on the horizontal axis.

(a)

What is the name of the shape formed by the scattered points? Use specific terms from portfolio theory.

Solution

The scattered portfolios form the opportunity set, which is the convex set of feasible risky portfolios in the (σ_Π, μ_Π) plane. The boundary of this set is the minimum-variance frontier, while the upper part of the boundary forms the efficient frontier (see [1], p. 78).

(b)

Identify the approximate coordinates

$$(\sigma_\Pi^{\min}, \mu_\Pi^{\min})$$

of the leftmost point of the boundary. What is this portfolio called?

Solution

The leftmost point of the opportunity set corresponds to the Global Minimum Variance (GMV) portfolio, since it is the feasible portfolio with the smallest possible portfolio standard deviation.

As shown in Figure 1, the simulated leftmost point is approximately

$$(\sigma_\Pi^{\min}, \mu_\Pi^{\min}) \approx (0.048082375,\ 0.042668753),$$

or approximately

$$(4.81\%,\ 4.27\%).$$

This is reasonably close to the analytical GMV result obtained previously,

$$(\sigma_{\Pi}^{\min}, \mu_{\Pi}^{\min}) \approx (4.76\%,\ 4.22\%).$$

The difference is expected because the Monte-Carlo experiment contains only a finite number of randomly generated portfolios and therefore does not necessarily sample the exact GMV allocation.

(c)

Why do no points appear to the left of this boundary? Give the mathematical reason in one sentence.

Solution

In the previous task, we analytically minimised $\mathbf{w}^T\Sigma\mathbf{w}$ under the same fully-invested constraint $\mathbf{1}^T\mathbf{w} = 1$, giving the GMV portfolio as the absolute minimum-risk point; more generally, every point on the minimum-variance frontier gives the smallest attainable variance for its corresponding expected return, so any feasible point lying further to the left would have a lower variance and would contradict the definition of the minimum-variance frontier.

(d)

A portfolio lies on the upper portion of this boundary at

$$\sigma_{\Pi} = 10\%.$$

Another portfolio lies on the lower portion at the same

$$\sigma_{\Pi} = 10\%.$$

Which portfolio would a rational investor choose and why?

Solution

A rational investor would choose the portfolio on the upper portion of the boundary, which lies on the efficient frontier. Both portfolios have the same risk, $\sigma_{\Pi} = 10\%$, but the portfolio on the upper branch provides the higher expected return and therefore dominates the portfolio on the lower branch.

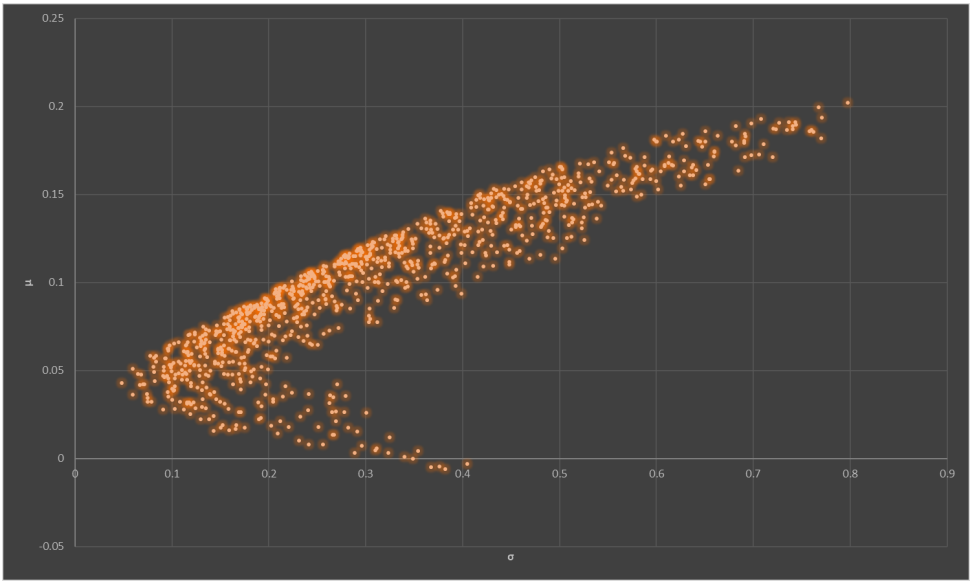


Figure 1: Monte-Carlo simulation of the feasible portfolio region.

B.Task – VaR Backtesting

The numerical work for this section is implemented in the accompanying Python notebook. The method follows the CQF backtesting procedure: daily log returns are used to estimate a rolling 21-day daily volatility, this is scaled to a 10-day horizon by the square-root-of-time rule, and the 99% Normal VaR threshold is then compared with the realised forward 10-day return (see [3], pp. 22–24).

For each date t,

$$\sigma_{10D,t} = \sqrt{10}\,\sigma_t,$$

and, with the mean return ignored as specified in the task,

$$\mathrm{VaR}_{10D,t} = -\Phi^{-1}(0.99)\sigma_{10D,t} \approx -2.32635\,\sigma_{10D,t}.$$

The realised forward 10-day log return is

$$r_{10D,t} = \ln\left(\frac{S_{t+10}}{S_t}\right),$$

and a breach is recorded when

$$r_{10D,t} < \mathrm{VaR}_{10D,t}.$$

The interpretation of VaR as a percentile threshold is reviewed in the CQF VaR notes (see [3], pp. 6–8).

(a) Breach count and percentage

The Python backtest gives

$$T = 622, \qquad b = 19, \qquad \frac{b}{T} = 3.05\%.$$

Therefore, 19 of the 622 eligible comparisons are VaR breaches. The observed breach rate is above the nominal 1% rate implied by a 99% VaR model.

(b) Visual identification of breaches

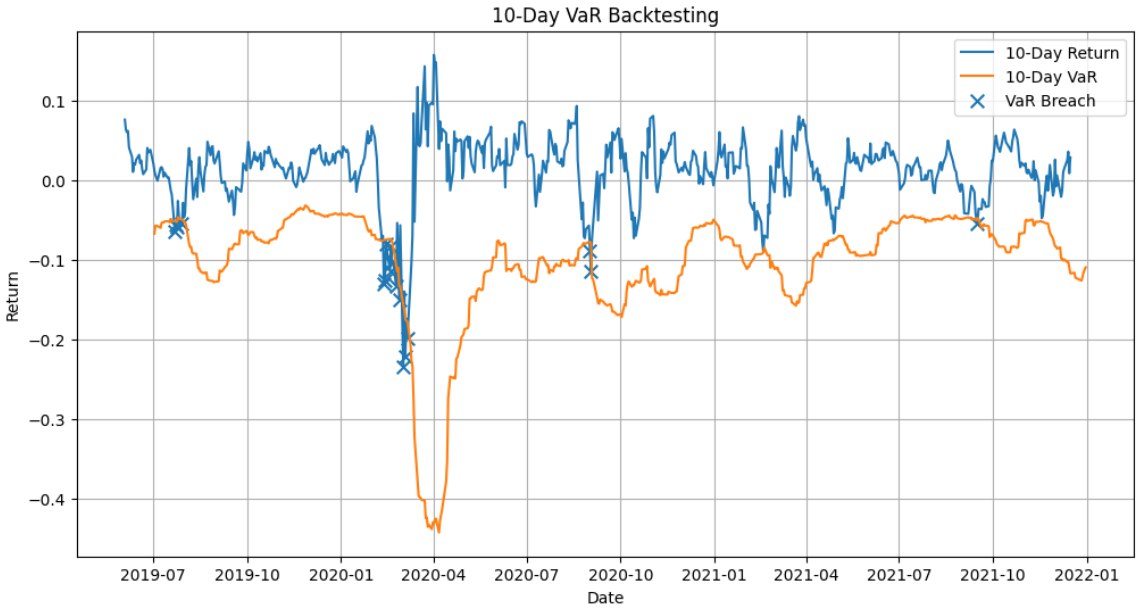


Figure 2: Rolling-SD 10-day VaR backtest. Crosses mark realised 10-day returns that fall below the VaR threshold.

(c) Traffic-light ranges

Using

$$B \sim \text{Binomial}(T, 0.01),$$

with $T = 622$, the Python calculation gives

$$q_G = 11, \qquad q_Y = 17.$$

Using the ranges specified in the task,

$$\begin{aligned} &\text{Green:} && 0 < b \leq 11, \\ &\text{Yellow:} && 11 < b \leq 17, \\ &\text{Red:} && b > 17. \end{aligned}$$

Since the observed number of breaches is $b = 19$, the rolling-SD VaR backtest falls in the red zone. In this sample the model therefore produces too many exceptions relative to the 99% VaR benchmark.

B.Task – EWMA Volatility Forecast and VaR

The backtest is repeated using an EWMA variance forecast with $\lambda = 0.72$. The dataset variance is used to initialise the recursion, after which

$$\sigma^2_{t+1|t} = \lambda\sigma^2_{t|t-1} + (1-\lambda)r_t^2.$$

The CQF EWMA material presents the same recursion and shows that the weights on past squared returns decay geometrically as

$$\alpha_i = (1-\lambda)\lambda^{i-1}, \qquad i \geq 1$$

(see [4], pp. 2–4). The resulting daily EWMA volatility is again scaled by $\sqrt{10}$ and converted to a 99% 10-day VaR using the same Normal factor.

(a) Breach count and percentage

The EWMA backtest gives

$$T = 642, \qquad b = 30, \qquad \frac{b}{T} = 4.67\%.$$

The EWMA model therefore produces 30 breaches in the eligible sample. Its breach percentage is higher than the rolling-SD result in this particular dataset. This does not mean that EWMA is less adaptive: adaptivity describes how quickly the volatility forecast responds to new information, whereas the breach rate measures calibration of the resulting VaR threshold.

(b) Visual identification of breaches

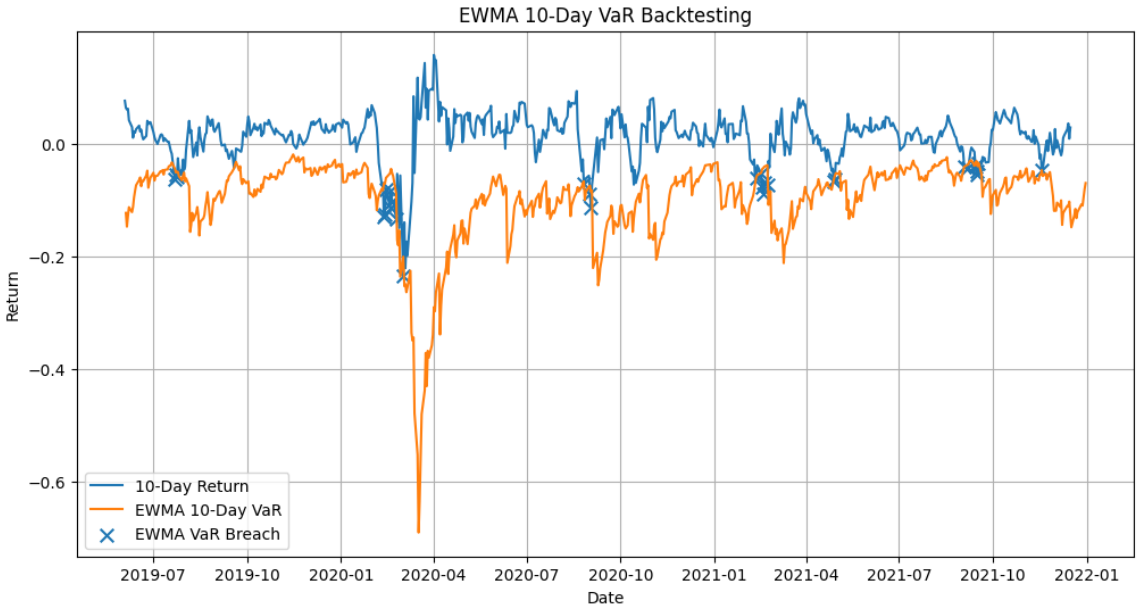


Figure 3: EWMA 10-day VaR backtest. Crosses mark realised 10-day returns that fall below the EWMA VaR threshold.

(c) Adaptivity of Rolling SD, EWMA and Historical Simulation

Rolling standard deviation

The rolling standard deviation estimates volatility from a fixed window of recent returns. In this task the window contains 21 daily returns. A large new return enters the estimate immediately, but it is averaged together with the other observations in the window. A shorter window is more responsive but noisier, while a longer window is smoother but slower to react. When an observation finally leaves the rolling window, its weight drops from an equal window weight to zero. The 21-day rolling construction used in the CQF backtesting example is described in [3], pp. 22–23.

EWMA

Among the three methods, EWMA has the strongest adaptivity to recent volatility in the form studied here. With $\lambda = 0.72$, the newest squared return receives weight

$$1 - \lambda = 0.28.$$

A large return therefore affects the next variance forecast immediately. Its influence then decays smoothly through the weights

$$(1-\lambda), \quad \lambda(1-\lambda), \quad \lambda^2(1-\lambda), \ldots$$

rather than remaining at a constant weight and then disappearing at a window boundary. The role of λ and this exponential decay are discussed in [4], pp. 2–4. The CQF VaR notes also discuss responsiveness of volatility estimation in backtesting (see [3], p. 24).

Historical simulation

Equal-weighted Historical Simulation has the weakest adaptivity of the three methods in the cases studied. It uses the empirical distribution of returns in a selected historical window, and the observations in that window receive equal weight. The VaR is determined by observations in the tail of that empirical distribution. It can therefore remain unchanged until an important extreme observation enters or leaves the window, producing characteristic plateaux. The CQF Historical Simulation notes show this behaviour directly and note the slow response of a long equal-weighted historical window to regime changes (see [3], pp. 11–13).

Overall, EWMA responds most smoothly and quickly to recent volatility changes, Rolling SD is intermediate and depends strongly on its chosen window length, and equal-weighted Historical Simulation can be the slowest to adapt.

CQF Material Referenced

The following CQF material was used to check terminology and methods in the report:

- **Fundamentals of Optimization and Application to Portfolio Selection** (annotated CQF lecture notes): pp. 14–17 for construction of the covariance matrix from standard deviations and correlations; pp. 78–84 for the minimum-variance frontier and GMV portfolio; p. 94 for the tangency-portfolio closed-form allocation; p. 108 for issues with historical expected-return estimates.
- **An Introduction to Portfolio Theory** (annotated CQF lecture notes): pp. 47–48 for the Capital Market Line, Sharpe ratio and tangency portfolio.
- **Notes on Statistical Essentials for Value at Risk and Expected Shortfall** (CQF teaching/tutorial notes, July 2026): pp. 6–8 for VaR as a percentile and the Normal factor; pp. 11–13 for Historical Simulation and its plateau behaviour; pp. 22–24 for rolling-volatility VaR backtesting, 10-day scaling and responsiveness.
- **Exponential Smoothing on Variance. ARCH Framework** (CQF Module Two supporting material): pp. 2–4 for the EWMA recursion, the role of λ, and exponentially declining weights.

References

[1] CQF, *Fundamentals of Optimization and Application to Portfolio Selection*, annotated lecture notes, Module 2, 2026.

[2] CQF, *An Introduction to Portfolio Theory*, annotated lecture notes, Module 2, 2026.

[3] Richard Diamond, *Notes on Statistical Essentials for Value at Risk and Expected Shortfall*, CQF teaching version, July 2026.

[4] CQF, *Exponential Smoothing on Variance. ARCH Framework*, Module Two supporting material, 2026.